GSK- IE Biopharma & AI Gateway
Clinical and Medical Research Challenges:
RWD TrustChain

Project Type:
Technological Architecture / Data Validation

GSK Mentor:
Javier Varas
Data Generation Director

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RWD TrustChain

Assigned GSK participant: Manu Vinuesa

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1. Introduction

Digital transformation in clinical research has created extensive data from sources like Electronic Health Records (EHRs), lab systems, and wearable devices. While offering valuable insights, it presents challenges in data integrity, security, and interoperability, which are crucial for compliance, scientific credibility, and patient safety.

2. The Problem

Real-world data (RWD) in fragmented systems can lead to duplication, inconsistencies, and data loss, reducing evidence quality. Manual entry and diverse platforms increase errors and weaken audit trails, while ensuring provenance, authenticity, and traceability becomes more complex but essential. In addition, Cybersecurity and regulations like GDPR and FDA 21 CFR Part 11 require organizations to balance collaboration with strict data protection.

3. The Challenge

Design a robust, secure, and interoperable solution that ensures:

- End-to-end traceability of clinical data across multiple sources.
- Validation and authentication of datasets using advanced technologies such as AI or blockchain.
- Streamlined RWD integration to support reliable Real-World Evidence generation.

4. The Prototype: “RWD TrustChain”

The RWD TrustChain is a conceptual design of a hybrid blockchain–AI infrastructure ensuring data traceability, security, and validation across decentralized clinical systems. The prototype features a flow diagram of the data lifecycle, a simulated blockchain ledger for data verification, and a dashboard monitoring data quality metrics that ensure the data lake would be use for Trials and RWE purposes.

Students should approach the problem by combining technological architecture, regulatory compliance, and methodological rigor—proposing a scalable framework applicable to multinational research settings.

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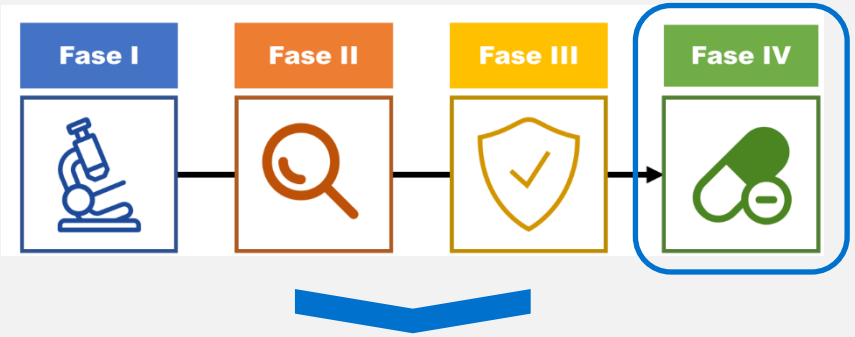
Introduction

Some definitions before starting the challenge:

- RWD
- RWE



- **Real World Data** → Includes all information related to a patient's health or their medical care in routine clinical practice and, therefore, collected outside the framework of a clinical trial.

- **Real World Evidence** → RWD organized and analyzed in such a way that, following the scientific method, allows for drawing conclusions.

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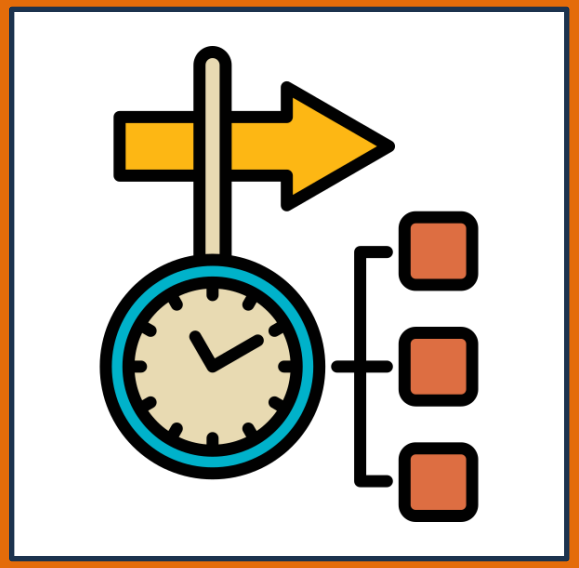
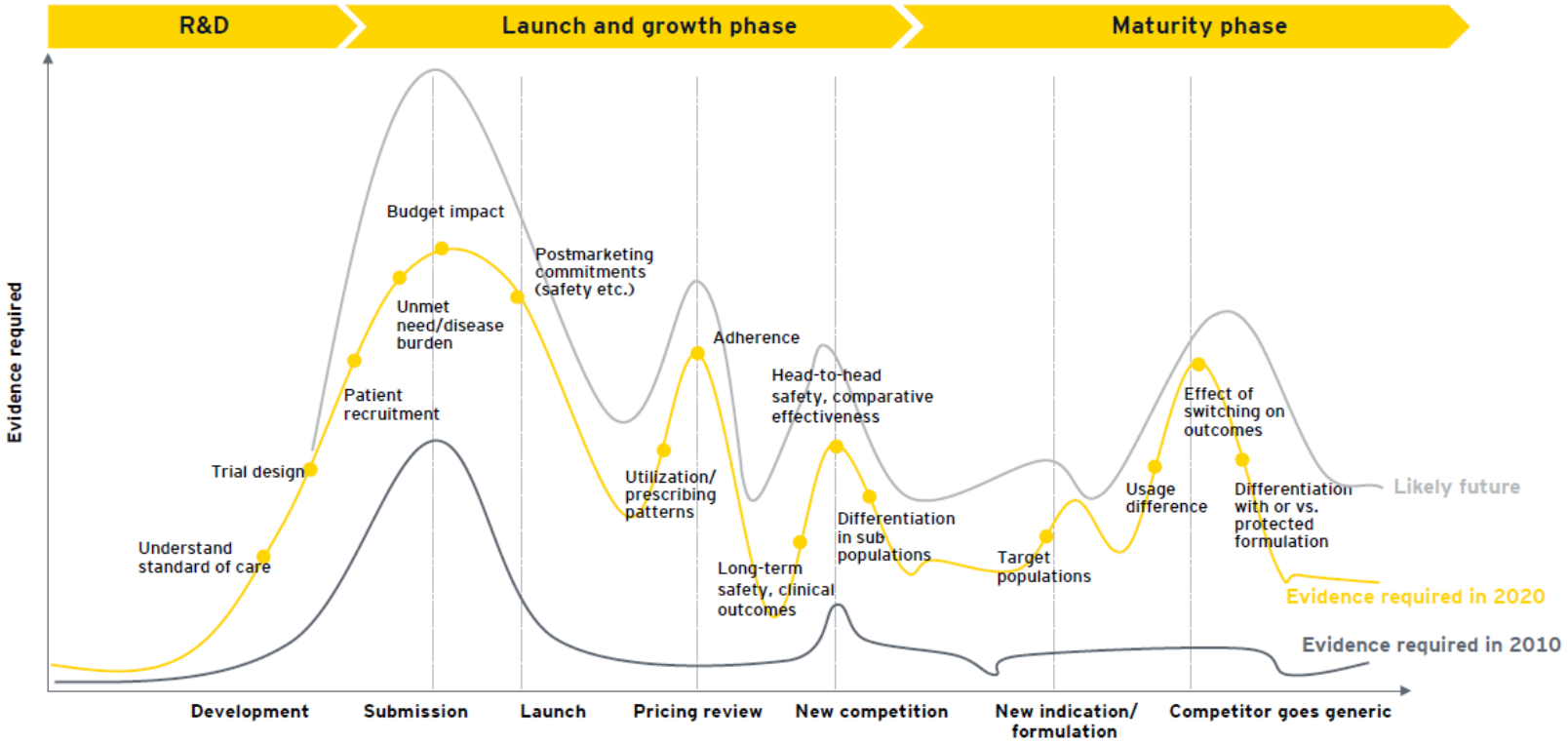
Introduction
Ok but… why is important the RWE and RWD?



Increasing evidence requirements across the drug lifecycle in past 10 years and the predicted trend for the future



What are the real-world evidence tools and how can they support decision making? EMA-EuropaBio Info Day – 22nd Nov 2016 Dr Alison Cave

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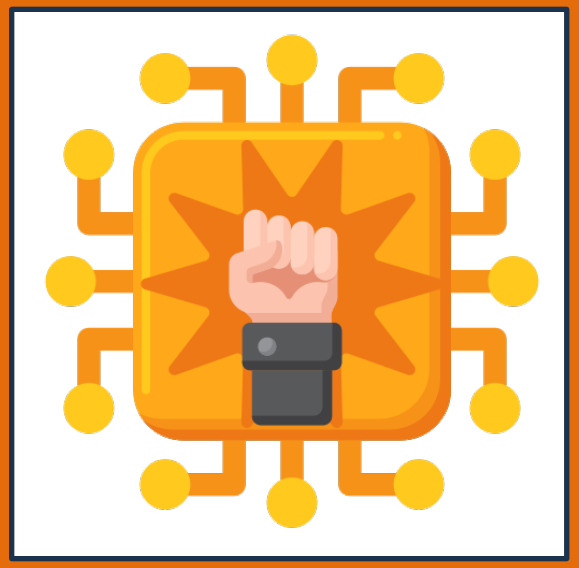
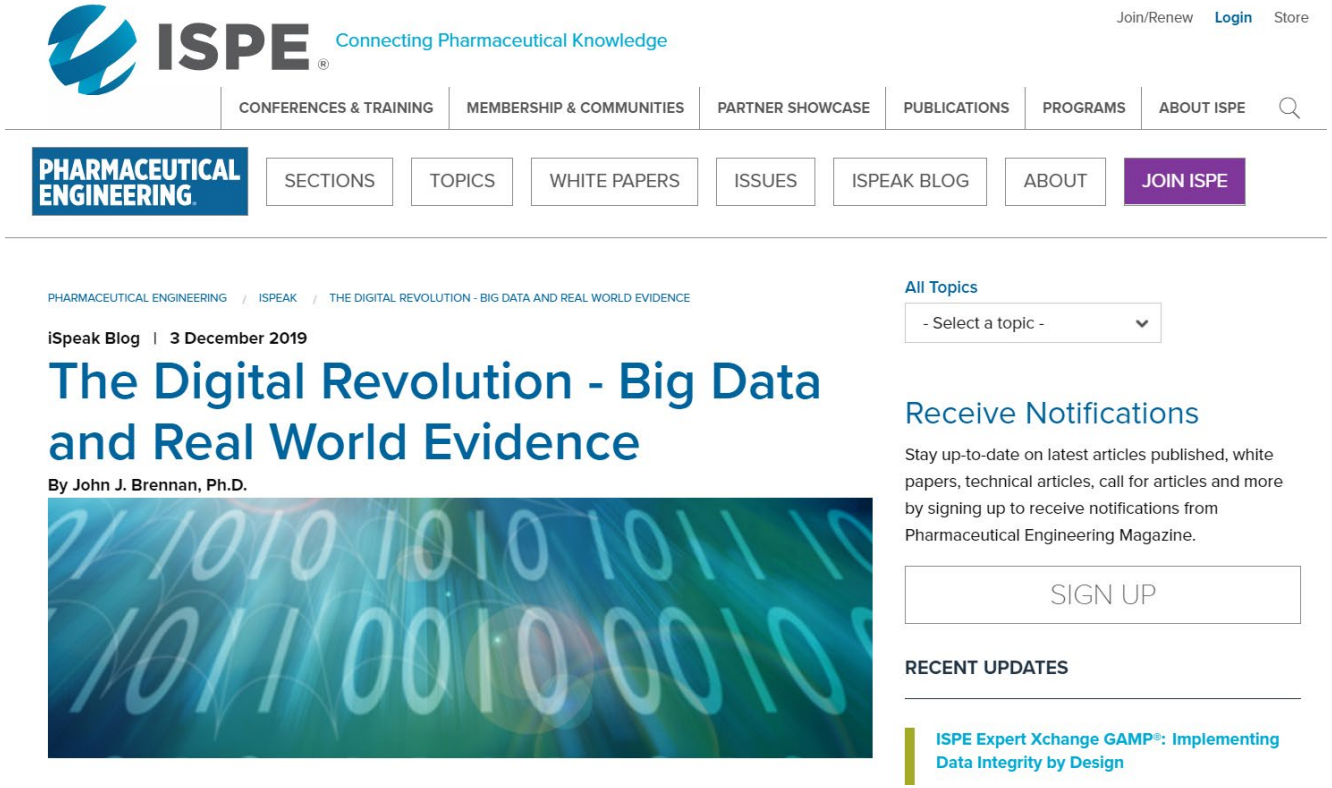


There is no doubt that we are witnessing a digital revolution in which Big Data generates powerful sources of RWD.



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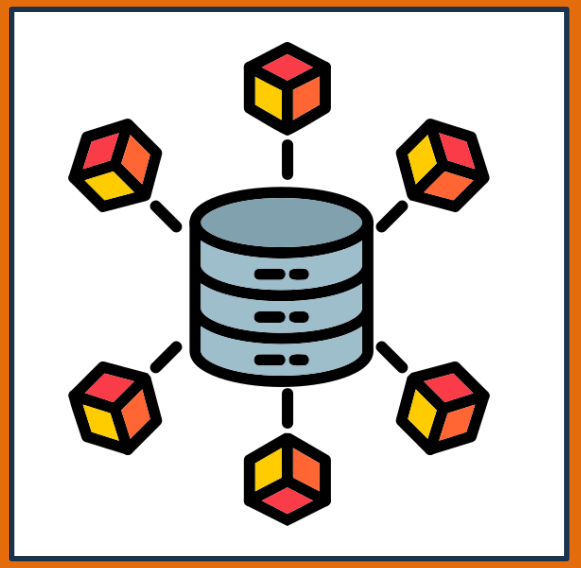
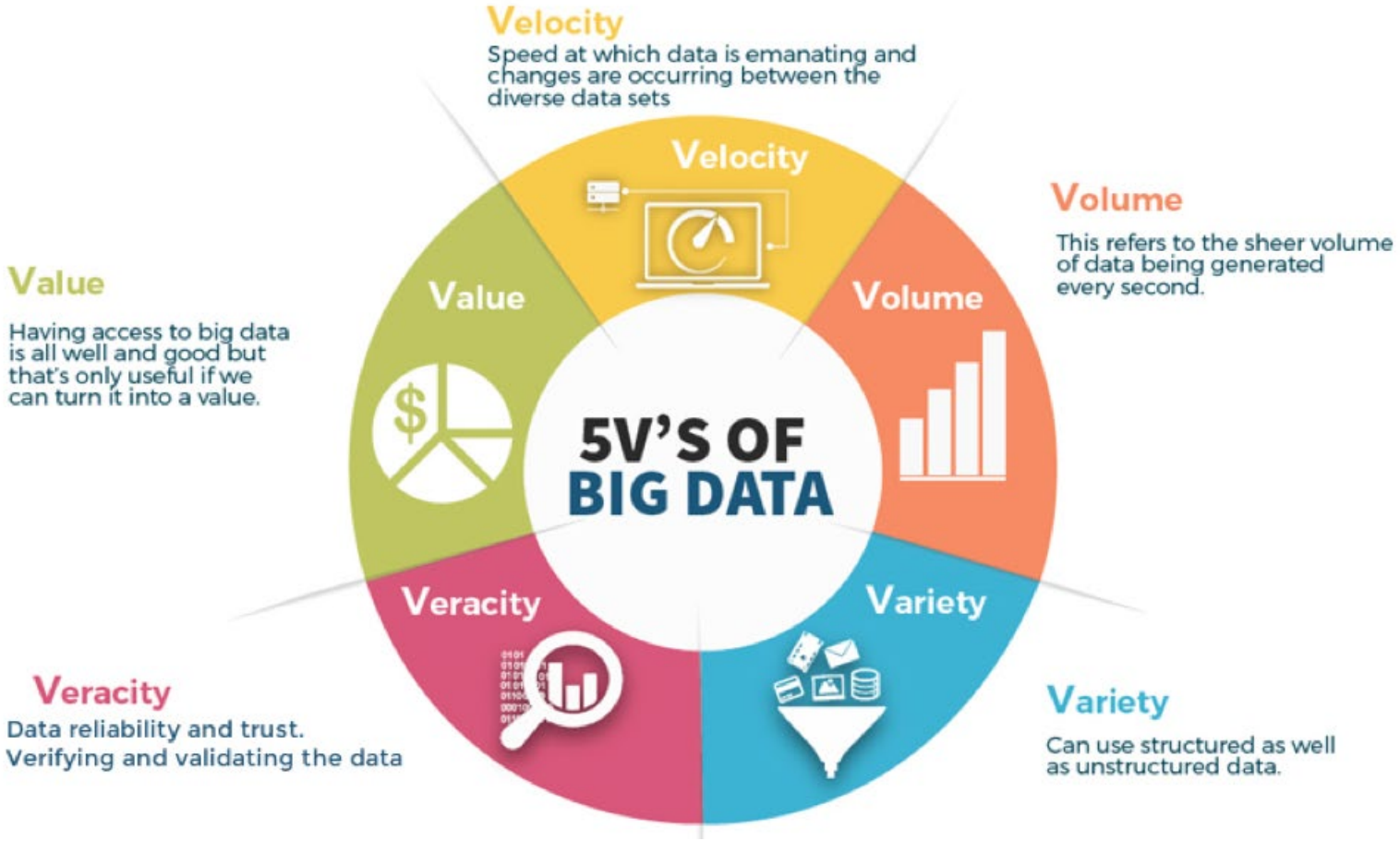


The 5V of Big Data



Rehman, Arshia & Naz, Saeeda & Razzak, Imran. (2021). Leveraging big data analytics in healthcare enhancement: trends, challenges and opportunities. Multimedia Systems. 28. 1-33. 10.1007/s00530-020-00736-8.

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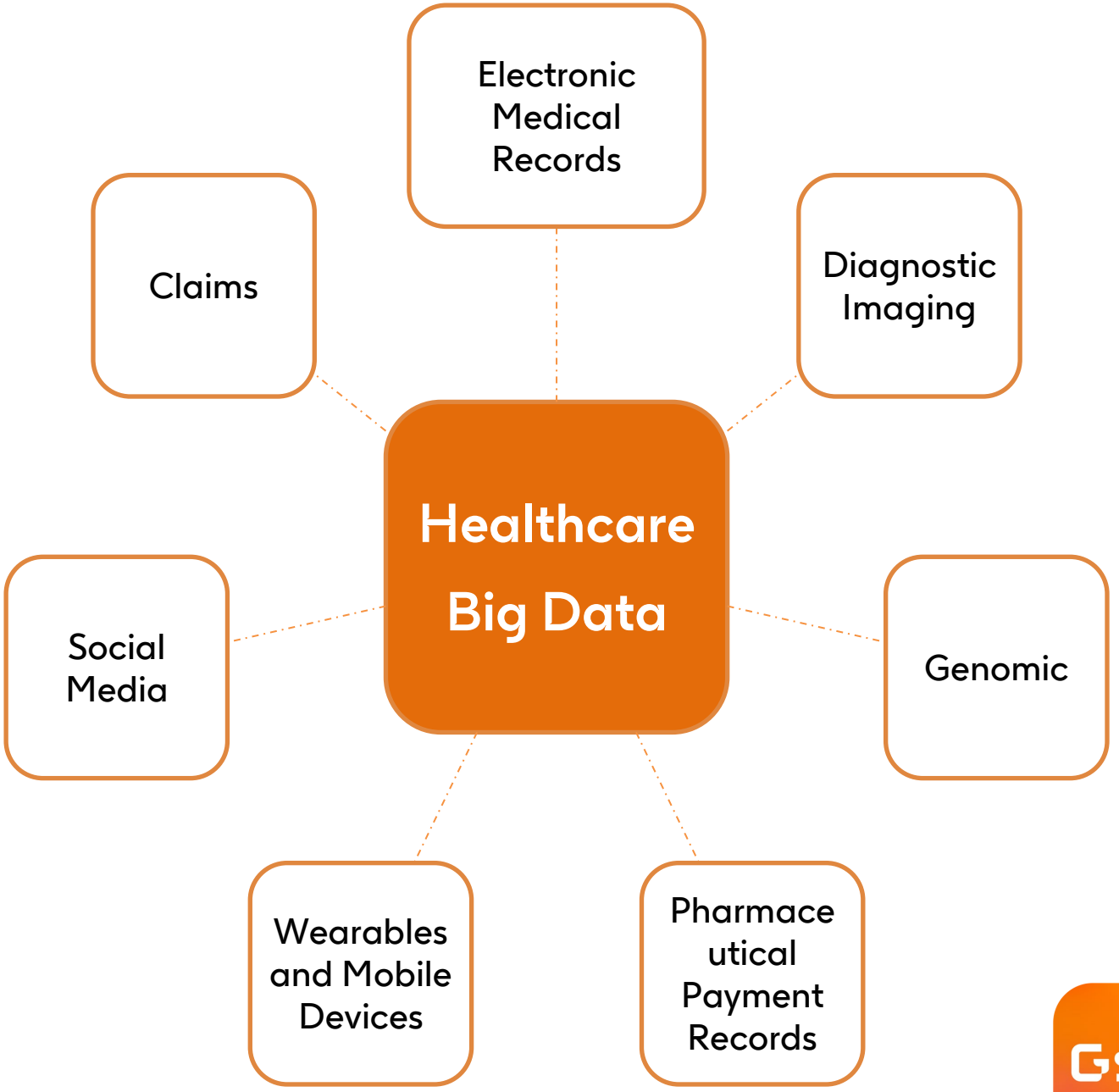
The Problem

1. Too many and unconnected sources



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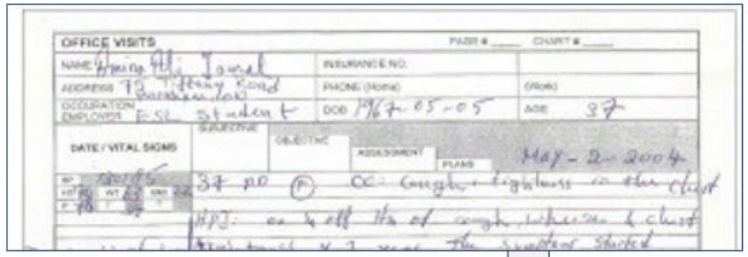
The Problem

2. Lack of structured data

Datos estructurados (≈20%)

Datos No estructurados (≈80%)

	registro	area	f_nac	edad	grupedad	peso	talla	imc
1	26	10	11572156800	50	50 - 54	71.4	152.0	30.90
2	32	10	11287728000	59	55 - 59	101.0	156.0	41.50
3	51	10	11154844800	63	60 - 64	74.0	165.0	27.18
4	144	10	11447568000	53	50 - 54	67.0	156.0	27.53
5	164	13	11394172800	55	55 - 59	70.0	142.0	34.72
6	182	10	11232086400	60	60 - 64	83.0	154.0	35.00
7	189	13	11133936000	64	60 - 64	67.0	160.0	26.17



Statistical Analysis

Final Study Report

Scientific Publication

[Hasta el 80% de los datos que provienen de las Organizaciones Sanitarias son datos no estructurados - Sedisa]

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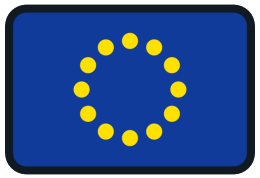
The problem

3 Data Protection regulation law asymmetries

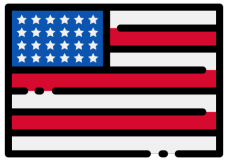
- **GDPR (General Data Protection Regulation)**

A comprehensive EU regulation that establishes strict requirements for how personal data—especially sensitive health information—must be collected, processed, stored, and shared. GDPR empowers individuals with strong data rights (such as access, portability, and erasure) and obliges organizations to implement robust security, transparency, and accountability measures. Its core principle is protecting individual privacy while enabling the responsible use of data for innovation and public benefit.



- **FDA 21 CFR Part 11**

A U.S. regulatory standard that governs the use of electronic records and electronic signatures in clinical trials, pharmaceutical manufacturing, and other regulated activities. It ensures data integrity, traceability, and reliability by requiring system validation, secure audit trails, controlled access, and trustworthy electronic approvals. Part 11 enables digital collaboration while guaranteeing that electronic data are as credible and compliant as paper records.



In an increasingly data-driven healthcare ecosystem, governments, hospitals, insurers, and pharmaceutical companies must advance collaboration without ever compromising the trust that comes from rigorous data protection

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